Lab 1: Pneumonia Classification Report

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1. Introduction

The objective of this experiment is to perform binary classification on chest X-ray images to determine whether a patient's lungs are "Normal" or indicate "Pneumonia". To achieve this, a custom PyTorch DataLoader was implemented to handle the Kaggle Chest X-Ray dataset by Paul Mooney (https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia). Two distinct deep residual network architectures, ResNet18 and ResNet50, were trained and evaluated to compare their feature extraction capabilities, accuracy, and F1-score trends.

2. Experiment Setups

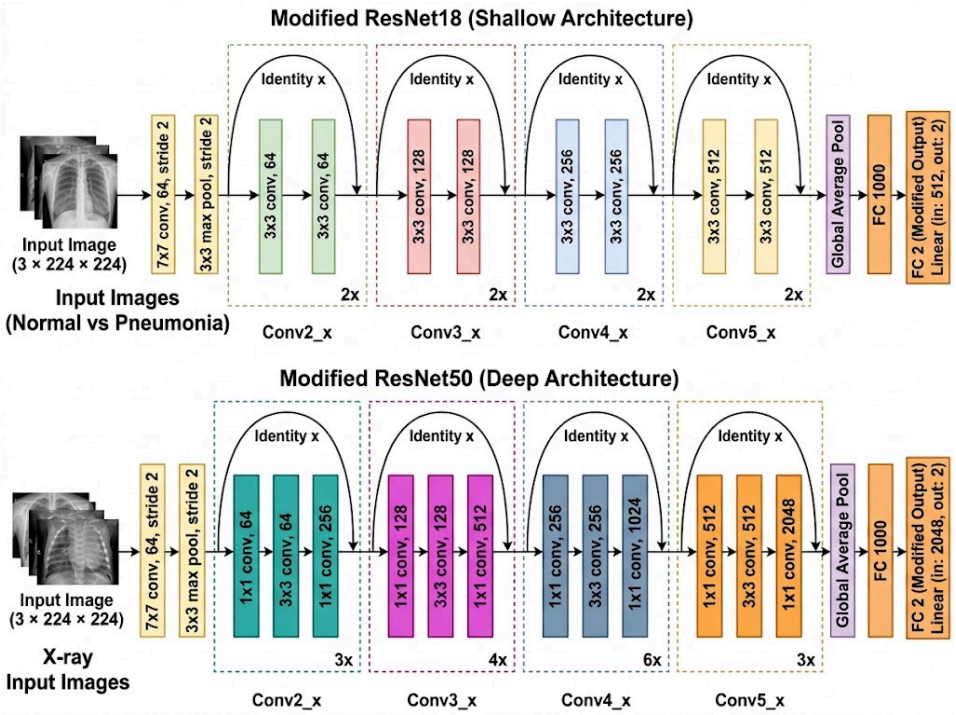


Figure 1: Architecture of Modified Resnet18 and Resnet50. The figure was generated using Gemini.

a. The Detail of the Model

Both models utilized ImageNet pre-trained weights from the PyTorch torchvision.models API. The final fully connected classification layer (model.fc) was re-initialized to output 2 classes instead of 1000. Here is the specification detail of both architectures:

Architectural Stage	**ResNet18 (Shallow)**	**ResNet50 (Deep)**
Input	3 × 224 × 224	3 × 224 × 224
Initial Layer	7 × 7 Conv, MaxPool	7 × 7 Conv, MaxPool
Blocks	Basic Blocks (2, 2, 2, 2)	Bottleneck Blocks (3, 4, 6, 3)
Pooling	Global Average Pooling (1x1)	Global Average Pooling (1x1)
Pre-trained Out	Linear (in: 512, out: 1000)	Linear (in: 2048, out: 1000)
Modified Out	**Linear (in: 512, out: 2)**	**Linear (in: 2048, out: 2)**

- **Loss Function:** To handle the severe class imbalance in the dataset (1,341 Normal vs. 3,875 Pneumonia), a weighted CrossEntropyLoss was applied with class weights [3.8896, 1.346]. This weight is based on the sample code and the data distribution.
- **Optimizer:** Adam optimizer was used with a learning rate of 1e-5. The weight decay was set to 0.9 for ResNet18 (10 epochs) and 1e-4 for ResNet50 (16 and 30 epochs) to manage overfitting.

b. The Detail of the DataLoader

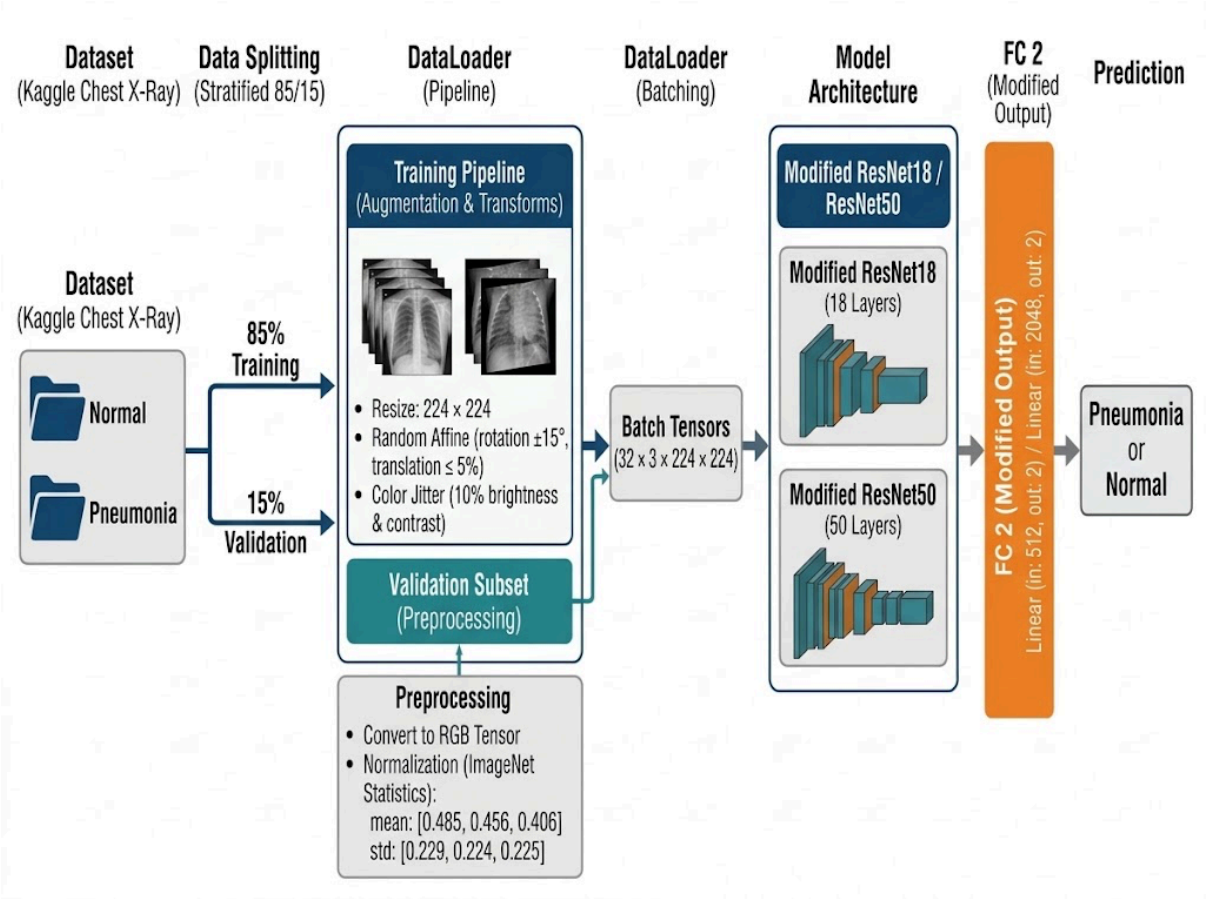


Figure2: A custom ChestXRayDataset class was written to load the images dynamically. The figure is generated with Gemini.

- **Data Split:** The base training dataset of 5,216 images was re-split into a stratified 85% Train (4,433 samples) and 15% Validation (783 samples) set to generate legitimate validation curves. The test set contained 624 images. The reason why I combined the validation is because in the original file it only has 16 images in total, 8 normal labels and 8 Pneumonia labels. Which will cause a massive swing in the validation process.
- **Data Augmentation:** The images were resized to 224x224 and converted to RGB tensors to follow the format input of Resnet architecture. The training transforms included RandomAffine (degrees -15 to +15, translate 5%) to simulate natural variations in patient positioning without introducing anatomically impossible flips that will introduce another medical condition such as Situs Inversus, and ColorJitter (brightness 0.1, contrast 0.1) to simulate varying X-ray exposures. Standard ImageNet normalization with mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225] was applied.

3. Experiment Result

a. Highest Testing Accuracy and F1-score

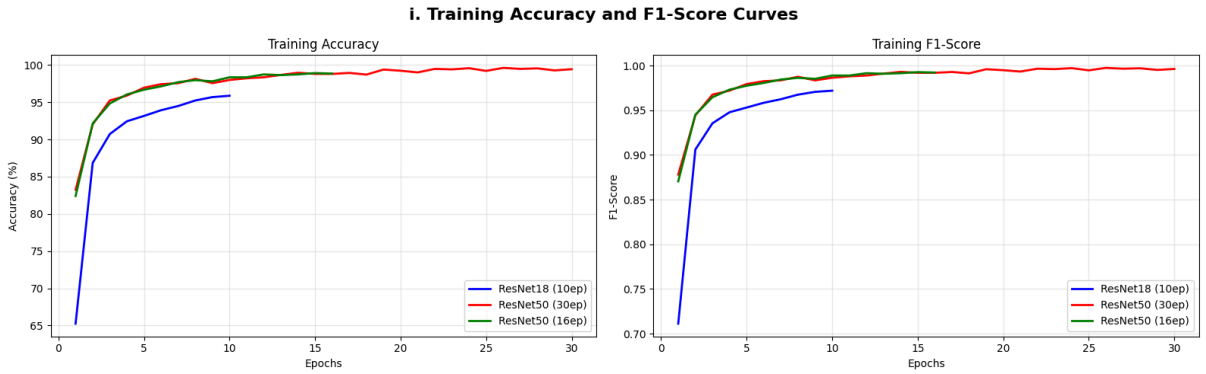
The ResNet18 architecture achieved the highest and most stable performance on the final test set.

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Test Acc: 92.31% | Recall: 0.9692 | Precision: 0.9130 | F1: 0.9403
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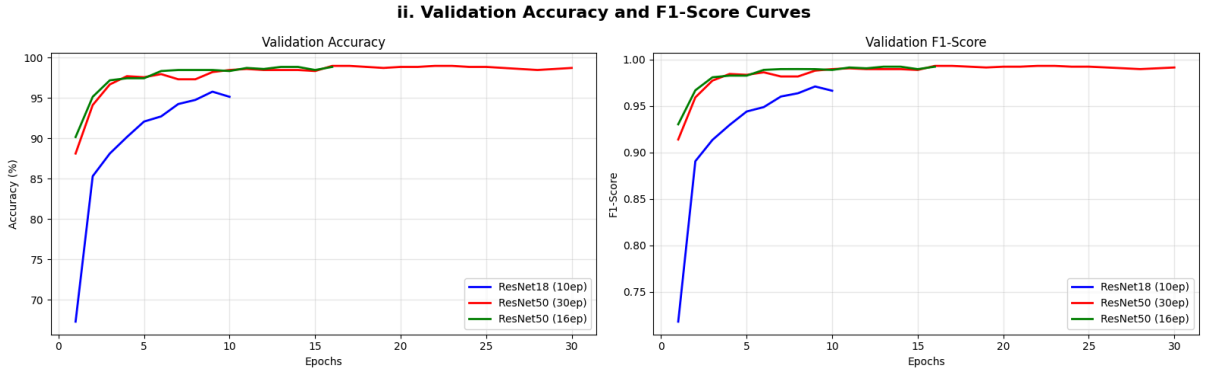
- **Test Accuracy:** 92.31%
- **F1-Score:** 0.9403
- **Recall:** 0.9692
- **Precision:** 0.9130

b. Plotting the Comparison Figure

i. Training accuracy and F1-score curves



ii. Validation accuracy and F1-score curves



iii. Highest Testing Accuracy Heatmap (Confusion Matrix)

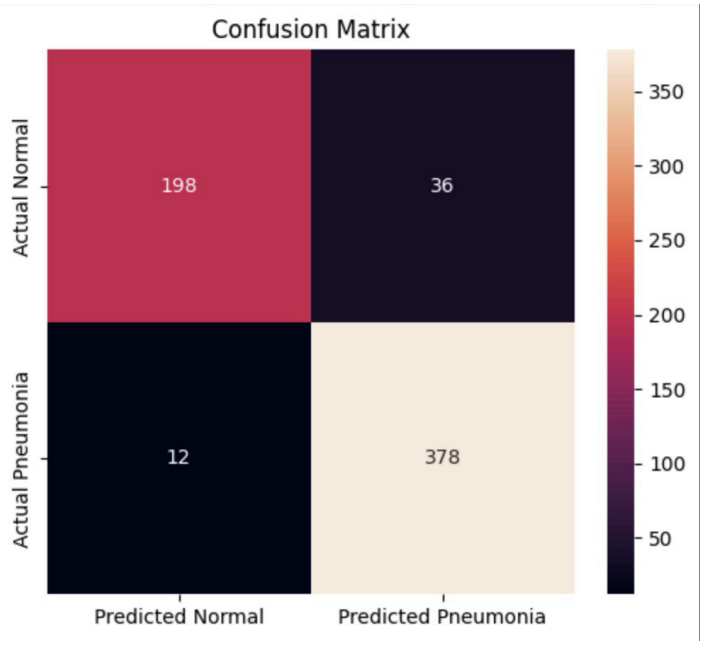


Figure 3: Confusion Matrix of Resnet18 with 10 Epochs

c. Training and validation Accuracy Curve & Validation F1-score Curve

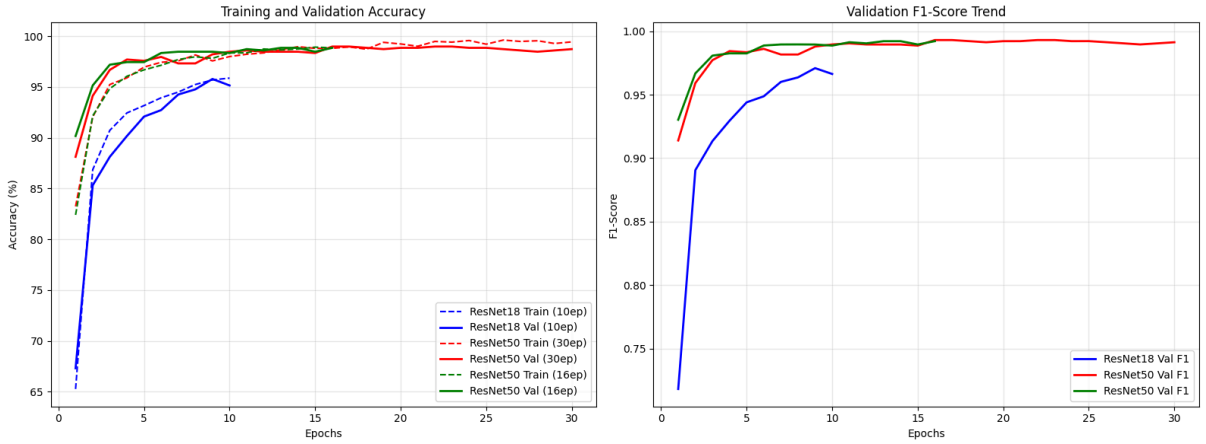


Figure 4: Training and Validation Curves of Both Architectures and Different Epoch

Resnet18 stopped at epoch 10 because it gained testing accuracy 92.31%. Resnet50 started with weight decay 0.9 which is too big for such more complex architecture so I changed to 1e-4. As we can see after epoch 16, Resnet50 training keeps on climbing but the validation starts to go down which signifies an overfitting that occurs.

4. Discussion

Architectural Trade-offs & Overfitting:

The comparison between ResNet18 and ResNet50 highlighted significant architectural trade-offs. While ResNet50 possesses a higher capacity for complex feature extraction, it began to overfit. The validation loss for ResNet50 reached its minimum at Epoch 16 (0.0369) before diverging and increasing in subsequent epochs, indicating the model began memorizing the training images. ResNet18 surprisingly generalized far better, maintaining a stable accuracy and a highly conservative 92.31% overall test accuracy. For Resnet50, I personally think that better data augmentation needs to be done in order to help it generalize. In some cases it catched the feature that is not in this model intended to do such as outliers or areas outside the chest cavity.

Impact of Class Weights:

Applying inverse frequency weights to the CrossEntropyLoss successfully helps the ResNet18 model yield a 0.9692 recall that is missing only 12 pneumonia cases out of 390. Minimizing False Negatives in medical diagnostics is the desired behavior.

Grad-CAM Analysis & The "Clever Hans" Effect:

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the final convolutional layer (layer4) for me to observe the model's decision-making process. This case I am using the model from Resnet18 and here are my findings:

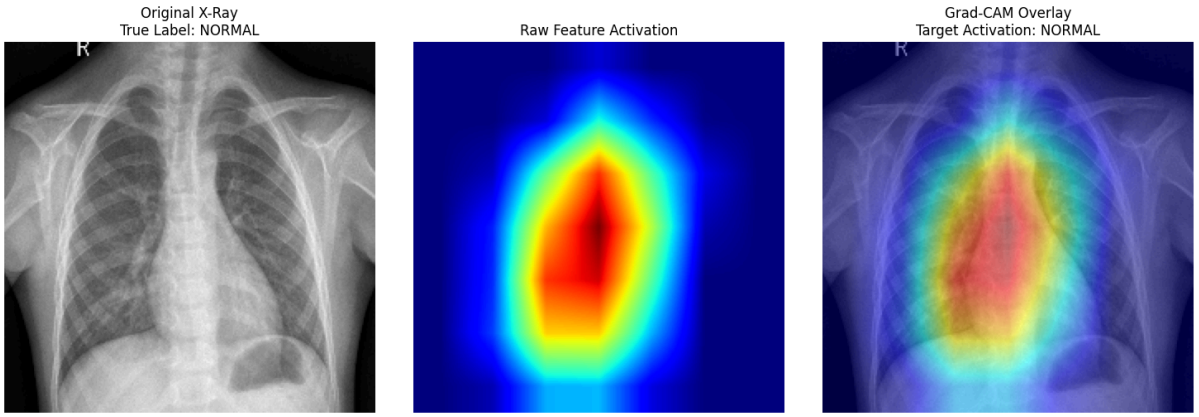


Figure 5: Normal Target Activation on Normal Label, Index 1

When targeting the "Normal" class, the model correctly focused its hot activation zones on the central mediastinum and clear lung fields.

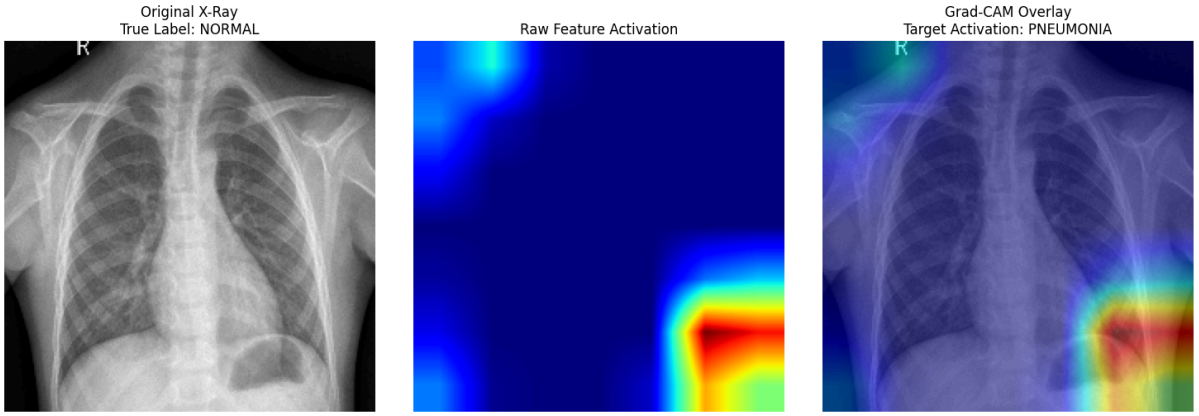


Figure 6: Normal Target Activation on Pneumonia Label, Index 1

However, when predicting "Pneumonia" on a “Normal” case, the Grad-CAM overlays revealed the model frequently relied on peripheral edge artifacts, such as the radiographic "R" marker and the patient's arm tissue rather than clinical pulmonary opacities. This reveals a "Clever Hans" effect, a term that I just learned from this case, where the network exploited framing shortcuts inherited from the dataset's source X-ray machines rather than relying purely on anatomical pathology.

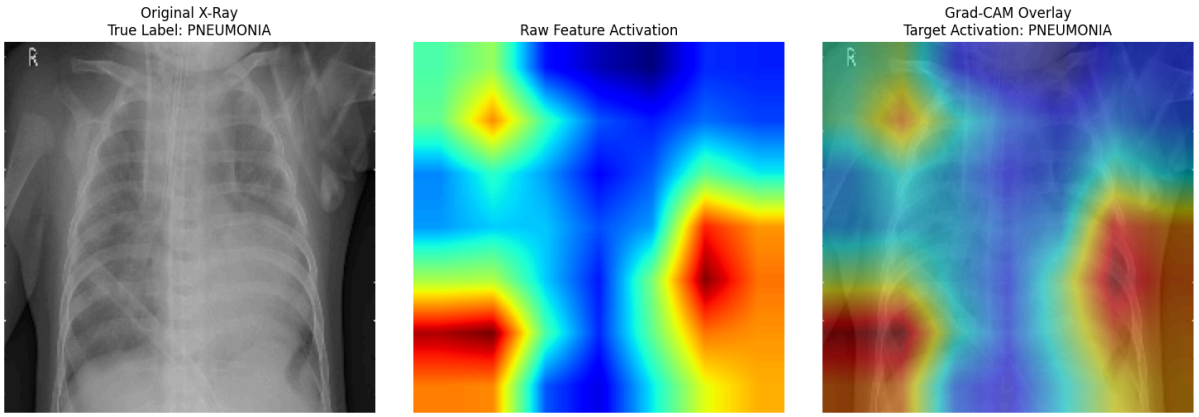


Figure 7: Pneumonia Target on Pneumonia Label, Index 234

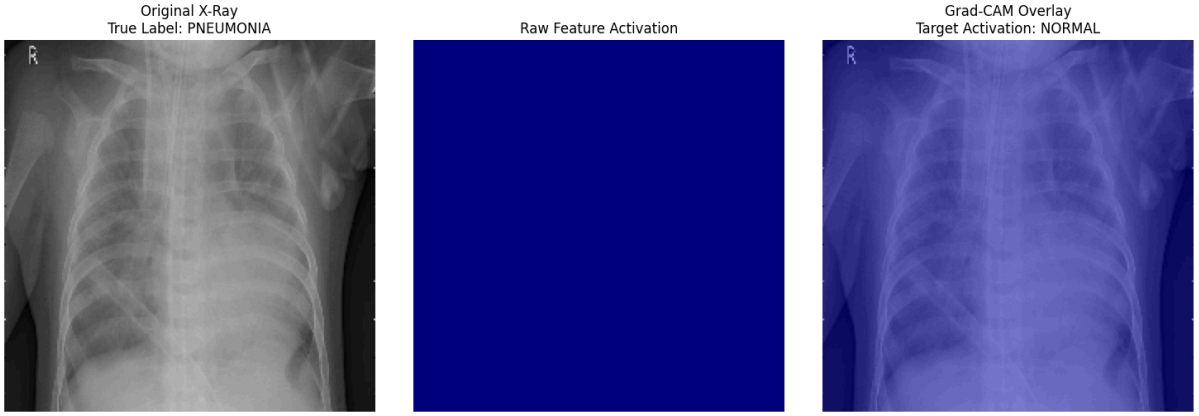


Figure 8: Normal Target Activation on Pneumonia Label, Index 234

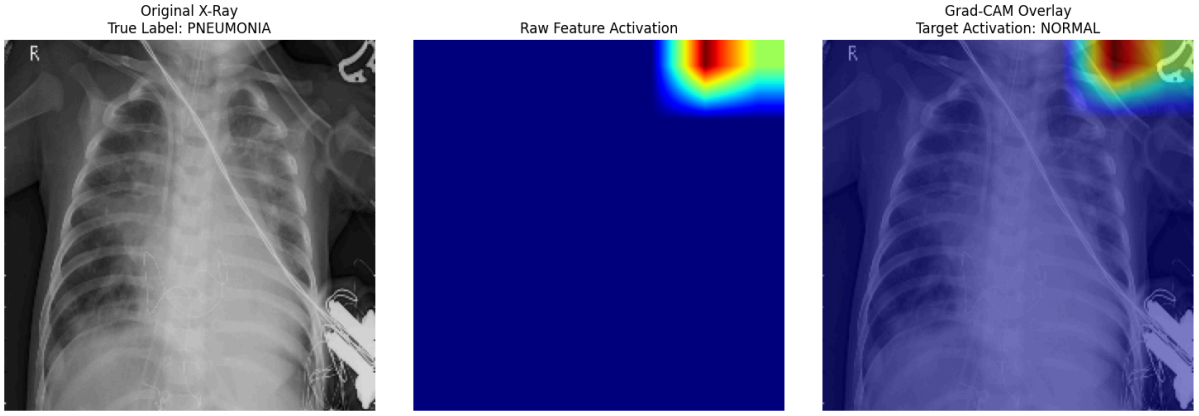


Figure 9: Normal Target Activation on Pneumonia Label, Index 236

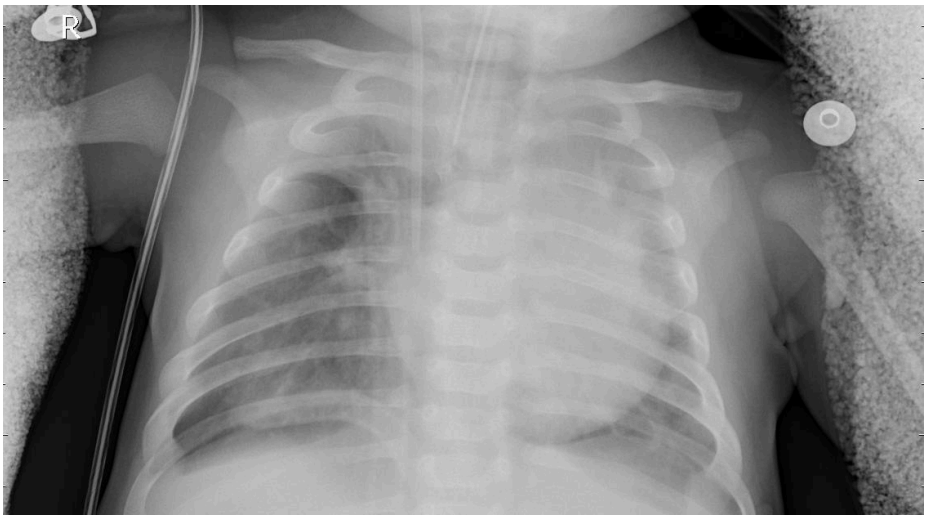


Figure 10: Outliers in Test Data “person100_bacteria_478.jpeg”

In Figure 7 the model could detect anomalies in the chest area and see it as a sign of Pneumonia. The Grad-CAM shows the model when detecting activation on Pneumonia data with Normal activation shows no region of interest in any chest area, this shows that model performs well in such cases. However, some flaw that model has is shown in Figure 9 where it detects the outliers rather than chest area. Even though the chest area is clean, this shows the model weakness in learning features. This drives further improvement in data preprocessing, and tells us that a strong architecture still needs a good quality of data.

5. Github Link

https://github.com/shem004/AIMI_Lab1_315551006_-.git